METALLURGY

1. Which element exists in free state

(1) Hg (2) Fe

(3) Au (4) Zn

2. Which statement is true

(1) All minerals are ore

(2) beneficial part of ore is gangue

(3) Among metal Fe is most abundant

(4) Sphalerite is the sulphide ore of Zn

3. Which one is not a ore of Iron?

(1) Siderite

(2) Iron pyrites

(3) Malachite

(4) Magnetite

4. Suitable ore for extraction of Fe is–

(1) Oxide ore

(2) Sulphide ore

(3) Carbonate ore

(4) all of these

5. Match the following

1. Zincite	P Sulphide ore
2. Malachite	Q halide ore
3. Horn silver	R Oxide ore
4. Iron pyrites	S Carbonate ore

(1) 1 – R; 2 – P; 3 – Q; 4 – S

(2) 1 – R; 2 – S; 3 – Q; 4 – P

(3) 1 – S; 2 – R; 3 – P; 4 – Q

(4) 1 – Q; 2 – S; 3 – P; 4 – R

6. Removal of unwanted material from ore is known as –

(1) Dressing of ore

(2) Benefaction of ore

(3) Concentration of ore

(4) all of these

7. In magnetic separation method which one is true

(1) either ore is being attracted by magnetic field

(2) either gangue is being attracted by magnetic field

(3) Both

(4) None

8. Froth stabilizer is–

(1) NaCN (2) Xanthate

(3) Cresol (4) pine oil

9. Which method is suitable for separation of ore from gangue, If ore is suitably soluble in some solvent.

(1) Leaching

(2) Poling

(3) Cupellation

(4) Zone refining method

10. Vapour phase refining method is applicable on

(1) Ni (2) Ti

(3) Zr (4) All

11. Column chromatographic separation method is based on

(1) Absorption (2) diffusion

(3) evaporation (4) Adsorption

12. Removal of different component from a adsorbent is known as–

(1) Leaching

(2) Elution

(3) Constant Boiling

(4) Vapour phase refining

13. Zinc is used for

(1) Hardening of steel

(2) galvanising the iron

(3) Softening of iron

(4) galvanising of Hg

14. In froth floatation method, concentrated ore is obtained at

(1) Bottom of container

(2) Skimmed off in form of upper layer

(3) Mixed in water-oil mixture

(4) Concentrated ore can't be obtained

15. Malleable iron is

(1) Pig iron (2) Cast iron

(3) spongy iron (4) wrought iron

16. Which reaction shows formation of blister copper?

(1) $2FeS + 3O_2 \longrightarrow 2FeO + 2SO_2\uparrow$

(2) $2Cu_2S + 3O_2 \longrightarrow 2Cu_2O + 2SO_2\uparrow$

(3) $2Cu_2O + Cu_2S \longrightarrow 6Cu + SO_2\uparrow$

(4) $Cu_2O + C \longrightarrow 2Cu + CO\uparrow$

17. If Na_2CO_3 is used to obtain alumina from Bauxite, process is known as–

(1) Halls method (2)Bayers method

(3) serpeck method (4) Halls-Herroult method

18. Which reaction is involved in extraction of Ag by cyanide process?

(1) $AgBr + Na_2S_2O_3 \longrightarrow Na_3[Ag(S_2O_3)_2]$

(2) $AgCl + NH_4OH \longrightarrow [Ag(NH_3)_2]Cl$

(3) $Ag_2S + NaCN \longrightarrow Na[Ag(CN)_2]_2$

(4) None

19. Impurity present in mineral is known as

(1) Flux (2) Slag

(3) gangue (4) All

20. Autoreduction process is used in metallurgy of

(1) Cu, Ag (2) Zn, Hg

(3) Cu, Hg (4) Zn, Ag

21. Bauxite is not purified by

(1) Halls process (2) Serpeck method

(3) Baeyer process (4) Hoop's method